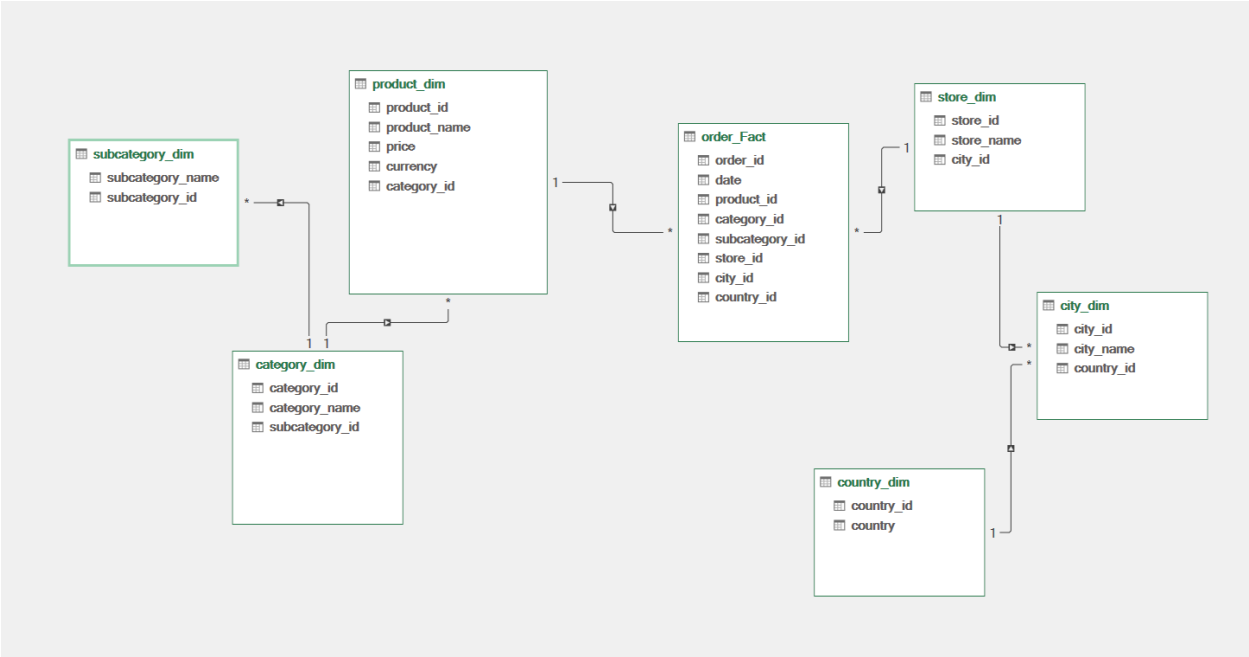
# Midterm Task 3 Normalization using Power Query

- This activity involves organizing and cleaning data using **Power Query** by applying normalization techniques. The goal is to transform unstructured or repeated data into a well-structured format that is easier to analyze and manage.


product_dim
product_id
product_name
price
currency
category_id
subcategory_dim
subcategory_name
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order_Fact
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date
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| A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|
| order_id | date | product_id | category_id | subcategory_id | store_id | city_id | country_id |
| CMT-36520 | 2/1/2020 | P002 | C01 | SC01 | S005 | MA | UK |
| CMT-36520 | 2/1/2020 | P020 | C03 | SC10 | S005 | MA | UK |
| CMT-36520 | 2/1/2020 | P012 | C02 | SC06 | S005 | MA | UK |
| CMT-36520 | 2/1/2020 | P012 | C02 | SC06 | S005 | MA | UK |
| CMT-36520 | 2/1/2020 | P005 | C01 | SC03 | S005 | MA | UK |
| MOF-12749 | 7/24/2020 | P004 | C01 | SC02 | S002 | LA | US |
| MOF-12749 | 7/24/2020 | P003 | C01 | SC02 | S002 | LA | US |
| MOF-12749 | 7/24/2020 | P016 | C03 | SC08 | S002 | LA | US |
| VKL-97451 | 12/19/2022 | P024 | C04 | SC12 | S005 | MA | UK |
| GJO-36640 | 4/14/2022 | P019 | C03 | SC10 | S005 | MA | UK |
| GJO-36640 | 4/14/2022 | P005 | C01 | SC03 | S005 | MA | UK |
| GJO-36640 | 4/14/2022 | P017 | C03 | SC09 | S005 | MA | UK |
| GJO-36640 | 4/14/2022 | P026 | C04 | SC13 | S005 | MA | UK |
| QNN-19269 | 10/2/2022 | P021 | C04 | SC11 | S001 | NY | US |
| QNN-19269 | 10/2/2022 | P023 | C04 | SC12 | S001 | NY | US |
| QNN-19269 | 10/2/2022 | P020 | C03 | SC10 | S001 | NY | US |
| QNN-19269 | 10/2/2022 | P011 | C02 | SC06 | S001 | NY | US |
| TFA-40365 | 10/25/2021 | P026 | C04 | SC13 | S005 | MA | UK |
| TFA-40365 | 10/25/2021 | P022 | C04 | SC11 | S005 | MA | UK |
| TFA-40365 | 10/25/2021 | P024 | C04 | SC12 | S005 | MA | UK |
| TFA-40365 | 10/25/2021 | P011 | C02 | SC06 | S005 | MA | UK |
| AQN-95165 | 1/16/2021 | P015 | C03 | SC08 | S003 | CH | US |
| PSG-64239 | 6/16/2020 | P001 | C01 | SC01 | S005 | MA | UK |
| PSG-64239 | 6/16/2020 | P020 | C03 | SC10 | S005 | MA | UK |
| PSG-64239 | 6/16/2020 | P016 | C03 | SC08 | S005 | MA | UK |
| OAT-63948 | 7/25/2023 | P018 | C03 | SC09 | S003 | CH | US |
| OAT-63948 | 7/25/2023 | P025 | C04 | SC13 | S003 | CH | US |
| IAY-77502 | 2/5/2020 | P026 | C04 | SC13 | S004 | LO | UK |
| IAY-77502 | 2/5/2020 | P005 | C01 | SC03 | S004 | LO | UK |
| IAY-77502 | 2/5/2020 | P024 | C04 | SC12 | S004 | LO | UK |
| IAY-77502 | 2/5/2020 | P003 | C01 | SC02 | S004 | LO | UK |
| IAY-77502 | 2/5/2020 | P014 | C02 | SC07 | S004 | LO | UK |
| JOH-53793 | 1/2/2023 | P013 | C02 | SC07 | S004 | LO | UK |
| JOH-53793 | 1/2/2023 | P017 | C03 | SC09 | S004 | LO | UK |
| JOH-53793 | 1/2/2023 | P003 | C01 | SC02 | S004 | LO | UK |
| JOH-53793 | 1/2/2023 | P002 | C01 | SC01 | S004 | LO | UK |
| JOH-53793 | 1/2/2023 | P016 | C03 | SC08 | S004 | LO | UK |
| OQS-86591 | 5/21/2021 | P018 | C03 | SC09 | S001 | NY | US |
| OQS-86591 | 5/21/2021 | P026 | C04 | SC13 | S001 | NY | US |

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | product_id | product_name | price | currency | category_id | |
| 2 | P002 | EconoTread | 699.99 | GBP | C01 | |
| 3 | P020 | Door Anchor Straps | 24.99 | GBP | C03 | |
| 4 | P012 | Olympic Flat Bench | 179.99 | GBP | C02 | |
| 5 | P005 | Glide Elite Elliptical | 999.99 | GBP | C01 | |
| 6 | P004 | Recumbent Cycler | 599.99 | USD | C01 | |
| 7 | P003 | SpinMaster Bike | 799.99 | USD | C01 | |
| 8 | P016 | Heavy Duty Resistance Tubes | 39.99 | USD | C03 | |
| 9 | P024 | Mini Percussion Massager | 79.99 | GBP | C04 | |
| 10 | P019 | Pro Suspension Trainer | 129.99 | GBP | C03 | |
| 11 | P017 | Cast Iron Kettlebell 16kg | 59.99 | GBP | C03 | |
| 12 | P026 | Extra Thick Exercise Mat | 49.99 | GBP | C04 | |
| 13 | P021 | Recovery Roller Deluxe | 39.99 | USD | C04 | |
| 14 | P023 | PulseRelief Massage Gun | 199.99 | USD | C04 | |
| 15 | P011 | Adjustable Bench Pro | 249.99 | USD | C02 | |
| 16 | P022 | Trigger Point Grid Roller | 59.99 | GBP | C04 | |
| 17 | P015 | FlexForce Band Kit | 49.99 | USD | C03 | |
| 18 | P001 | Pro Runner 3000 | 1299.99 | GBP | C01 | |
| 19 | P018 | Adjustable Kettlebell | 149.99 | USD | C03 | |
| 20 | P025 | Premium Yoga Mat | 69.99 | USD | C04 | |
| 21 | 014 | Leg Press Machine | 899.99 | GBP | C02 | |
| 22 | P013 | Multi-Gym Station | 1499.99 | GBP | C02 | |
| 23 | P007 | AquaRow 500 | 899.99 | USD | C01 | |
| 24 | P006 | Compact Cross Trainer | 549.99 | USD | C01 | |
| 25 | P008 | AirRow Pro | 1099.99 | USD | C01 | |
| 26 | P010 | Cast Iron Plate Set | 199.99 | GBP | C02 | |
| 27 | P009 | PowerLift Dumbbell Set | 299.99 | USD | C02 | |

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | category_id | category_name | subcategory_id | | | |
| 2 | C01 | Cardio Equipment | SC01 | | | |
| 3 | C03 | Functional Fitness | SC10 | | | |
| 4 | C02 | Strength Training | SC06 | | | |
| 5 | C04 | Recovery and Wellness | SC12 | | | |
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| 1 | subcategory_name | subcategory_id | | | | |
| 2 | Treadmills | SC01 | | | | |
| 3 | Suspension Trainers | SC10 | | | | |
| 4 | Weight Benches | SC06 | | | | |
| 5 | Ellipticals | SC03 | | | | |
| 6 | Exercise Bikes | SC02 | | | | |
| 7 | Resistance Bands | SC08 | | | | |
| 8 | Massage Guns | SC12 | | | | |
| 9 | Kettlebells | SC09 | | | | |
| 10 | Yoga Mats | SC13 | | | | |
| 11 | Foam Rollers | SC11 | | | | |
| 12 | Weight Machines | SC07 | | | | |
| 13 | Rowing Machines | SC04 | | | | |
| 14 | Free Weights | SC05 | | | | |
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| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | store_id | store_name | city_id | | | |
| 2 | S005 | Active Life Outfitters | MA | | | |
| 3 | S002 | Muscle Mart | LA | | | |
| 4 | S001 | FitGear Hub | NY | | | |
| 5 | S003 | Strength Central | CH | | | |
| 6 | S004 | Elite Gym Supply | LO | | | |
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| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | city_id | city_name | country_id | | | |
| 2 | MA | Manchester | UK | | | |
| 3 | LA | Los Angeles | US | | | |
| 4 | NY | New York | US | | | |
| 5 | CH | Chicago | US | | | |
| 6 | LO | London | UK | | | |
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| 10 | | | | | | |
| 11 | | | | | | |

| | A | B | C | D | E | F |
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| 1 | country_id | country | | | | |
| 2 | UK | United Kingdom | | | | |
| 3 | US | United States | | | | |
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https://docs.google.com/spreadsheets/d/1CZMiFfcIHA8iJ7nkGATnTE-JxwMnGeoe/edit?usp=sharing&ouid=117777993764817776816&rtpof=true&sd=true